H2: Determinant (uigewerkte oplossingen) Codeertechniek van Hill

```
(%i1)   rref(A):=block([p,q,k],[p,q]:matrix_size(A),A:echelon(A),
            k:min(p,q),
            for i thru min(p,q) do (if A[i,i]=0 then (k:i-1,return())),
            for i:k thru 2 step -1 do (for j from i-1 thru 1 step -1 do A:rowop(A,j,i,A[j,i])),
            A)$
```

1) Voorbeeld uit cursus

Sleutelmatrix

```
(%i2)   S: matrix(
         [2,1,2],
         [1,0,2],
         [3,1,5]
        );
```

(S) $\begin{bmatrix} 2 & 1 & 2 \\ 1 & 0 & 2 \\ 3 & 1 & 5 \end{bmatrix}$

```
(%i3)   determinant(S);
(%o3)   -1
```

Determinant verschillend van 0, dus een geschikte sleutelmatrix

(%i4)
```
B: matrix(
 [8,12],
 [1,15],
 [12,0]
);
```

(B) $$\begin{bmatrix} 8 & 12 \\ 1 & 15 \\ 12 & 0 \end{bmatrix}$$

(%i5)
```
S.B;
```

(%o5) $$\begin{bmatrix} 41 & 39 \\ 32 & 12 \\ 85 & 51 \end{bmatrix}$$

Gecodeerde boodschap: 41 32 85 | 39 12 51

Inverse sleutelmatrix

(%i6)
```
SI: matrix(
 [2,1,2,1,0,0],
 [1,0,2,0,1,0],
 [3,1,5,0,0,1]
);
```

(SI) $$\begin{bmatrix} 2 & 1 & 2 & 1 & 0 & 0 \\ 1 & 0 & 2 & 0 & 1 & 0 \\ 3 & 1 & 5 & 0 & 0 & 1 \end{bmatrix}$$

(%i7)
```
rref(SI);
```

0 errors, 0 warnings

(%o7) $$\begin{bmatrix} 1 & 0 & 0 & 2 & 3 & -2 \\ 0 & 1 & 0 & -1 & -4 & 2 \\ 0 & 0 & 1 & -1 & -1 & 1 \end{bmatrix}$$

```
(%i8)   Sinv:submatrix(%,1,2,3);
```

(Sinv) $\begin{bmatrix} 2 & 3 & -2 \\ -1 & -4 & 2 \\ -1 & -1 & 1 \end{bmatrix}$

Controle: Sinv^^(-1)=S

```
(%i9)   Sinv^^(-1);
```

(%o9) $\begin{bmatrix} 2 & 1 & 2 \\ 1 & 0 & 2 \\ 3 & 1 & 5 \end{bmatrix}$

```
(%i10)  invert(Sinv);
```

(%o10) $\begin{bmatrix} 2 & 1 & 2 \\ 1 & 0 & 2 \\ 3 & 1 & 5 \end{bmatrix}$

Ontcijferen:

```
(%i11)  C: matrix(
          [41,39],
          [32,12],
          [85,51]
        );
```

(C) $\begin{bmatrix} 41 & 39 \\ 32 & 12 \\ 85 & 51 \end{bmatrix}$

```
(%i12)  Sinv.C;
```

(%o12) $\begin{bmatrix} 8 & 12 \\ 1 & 15 \\ 12 & 0 \end{bmatrix}$

2) oef 26

Ingeven code:

```
(%i13) C: matrix(
         [-1,2,46,62,11],
         [-50,-59,-4,-5,-80],
         [31,36,-5,-7,48]
       );
```

(C) $$\begin{bmatrix} -1 & 2 & 46 & 62 & 11 \\ -50 & -59 & -4 & -5 & -80 \\ 31 & 36 & -5 & -7 & 48 \end{bmatrix}$$

Sleutelmatrix:

```
(%i14) S: matrix(
         [2,-1,2],
         [1,-3,-2],
         [-1,2,1]
       );
```

(S) $$\begin{bmatrix} 2 & -1 & 2 \\ 1 & -3 & -2 \\ -1 & 2 & 1 \end{bmatrix}$$

```
(%i15) determinant(S);
(%o15) -1
```

Inverse sleutelmatrix:

(%i16)
```
SI: matrix(
 [2,-1,2,1,0,0],
 [1,-3,-2,0,1,0],
 [-1,2,1,0,0,1]
);
```

(SI)
$$\begin{bmatrix} 2 & -1 & 2 & 1 & 0 & 0 \\ 1 & -3 & -2 & 0 & 1 & 0 \\ -1 & 2 & 1 & 0 & 0 & 1 \end{bmatrix}$$

(%i17)
```
rref(SI);
```

(%o17)
$$\begin{bmatrix} 1 & 0 & 0 & -1 & -5 & -8 \\ 0 & 1 & 0 & -1 & -4 & -6 \\ 0 & 0 & 1 & 1 & 3 & 5 \end{bmatrix}$$

(%i18)
```
Sinv:submatrix(%,1,2,3);
```

(Sinv)
$$\begin{bmatrix} -1 & -5 & -8 \\ -1 & -4 & -6 \\ 1 & 3 & 5 \end{bmatrix}$$

Controle:

(%i19)
```
invert(Sinv);
```

(%o19)
$$\begin{bmatrix} 2 & -1 & 2 \\ 1 & -3 & -2 \\ -1 & 2 & 1 \end{bmatrix}$$

Decoderen:

(%i20)
```
Sinv.C;
```

(%o20)
$$\begin{bmatrix} 3 & 5 & 14 & 19 & 5 \\ 15 & 18 & 0 & 0 & 21 \\ 4 & 5 & 9 & 12 & 11 \end{bmatrix}$$

Vertalen volgens rangnummer:

Coderen is leuk

3) oef 27

Ingeven code:

```
(%i21) C: matrix(
         [84,55,131,74,66,70,81],
         [40,25,63,37,33,25,39],
         [54,30,106,55,61,45,58]
       );
```

(C) $$\begin{bmatrix} 84 & 55 & 131 & 74 & 66 & 70 & 81 \\ 40 & 25 & 63 & 37 & 33 & 25 & 39 \\ 54 & 30 & 106 & 55 & 61 & 45 & 58 \end{bmatrix}$$

Sleutelmatrix:

```
(%i22) S: matrix(
         [3,4,2],
         [1,2,1],
         [2,4,1]
       );
```

(S) $$\begin{bmatrix} 3 & 4 & 2 \\ 1 & 2 & 1 \\ 2 & 4 & 1 \end{bmatrix}$$

```
(%i23) determinant(S);
(%o23) -2
```

Inverse sleutelmatrix:

(%i24)
```
SI: matrix(
 [3,4,2,1,0,0],
 [1,2,1,0,1,0],
 [2,4,1,0,0,1]
);
```

(SI) $$\begin{bmatrix} 3 & 4 & 2 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 0 \\ 2 & 4 & 1 & 0 & 0 & 1 \end{bmatrix}$$

(%i25)
```
rref(SI);
```

(%o25) $$\begin{bmatrix} 1 & 0 & 0 & 1 & -2 & 0 \\ 0 & 1 & 0 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 1 & 0 & 2 & -1 \end{bmatrix}$$

(%i26)
```
Sinv:submatrix(%,1,2,3);
```

(Sinv) $$\begin{bmatrix} 1 & -2 & 0 \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 0 & 2 & -1 \end{bmatrix}$$

Controle:

(%i27)
```
invert(Sinv);
```

(%o27) $$\begin{bmatrix} 3 & 4 & 2 \\ 1 & 2 & 1 \\ 2 & 4 & 1 \end{bmatrix}$$

Decoderen:

(%i28) `Sinv.C;`

(%o28) $$\begin{bmatrix} 4 & 5 & 5 & 0 & 0 & 20 & 3 \\ 5 & 0 & 19 & 9 & 14 & 0 & 8 \\ 26 & 20 & 20 & 19 & 5 & 5 & 20 \end{bmatrix}$$

Vertalen volgens rangnummer:

Deze test is net echt.

4) oef 28

4.1 (a)

Is M een geldige sleutelmatrix?

(%i29)
```
M: matrix(
  [1,14,14,4],
  [11,7,6,9],
  [8,10,10,5],
  [13,2,3,15]
);
```

(M) $$\begin{bmatrix} 1 & 14 & 14 & 4 \\ 11 & 7 & 6 & 9 \\ 8 & 10 & 10 & 5 \\ 13 & 2 & 3 & 15 \end{bmatrix}$$

(%i30) `determinant(M);`

(%o30) -1485

Besluit: det(M)!=0 dus M is een geldige sleutelmatrix -> de bewering is WAAR

4.2 (b)

(%i31)
```
C: matrix(
 [506,150,470,369],
 [482,390,521,419],
 [475,285,493,417],
 [484,495,562,412]
);
```

(C)
$$\begin{bmatrix} 506 & 150 & 470 & 369 \\ 482 & 390 & 521 & 419 \\ 475 & 285 & 493 & 417 \\ 484 & 495 & 562 & 412 \end{bmatrix}$$

(%i32)
```
MI:addcol(M,ident(4));
```

(MI)
$$\begin{bmatrix} 1 & 14 & 14 & 4 & 1 & 0 & 0 & 0 \\ 11 & 7 & 6 & 9 & 0 & 1 & 0 & 0 \\ 8 & 10 & 10 & 5 & 0 & 0 & 1 & 0 \\ 13 & 2 & 3 & 15 & 0 & 0 & 0 & 1 \end{bmatrix}$$

(%i33)
```
rref(MI);
```

(%o33)
$$\begin{bmatrix} 1 & 0 & 0 & 0 & -\frac{13}{99} & -\frac{2}{99} & \frac{20}{99} & -\frac{2}{99} \\ 0 & 1 & 0 & 0 & \frac{67}{495} & \frac{353}{495} & -\frac{263}{495} & -\frac{142}{495} \\ 0 & 0 & 1 & 0 & -\frac{43}{495} & -\frac{362}{495} & \frac{287}{495} & \frac{133}{495} \\ 0 & 0 & 0 & 1 & \frac{56}{495} & \frac{34}{495} & -\frac{109}{495} & \frac{34}{495} \end{bmatrix}$$

Controle:

(%i34) `Minv:M^^(-1);`

(Minv)
$$\begin{bmatrix} -\frac{13}{99} & -\frac{2}{99} & \frac{20}{99} & -\frac{2}{99} \\ \frac{67}{495} & \frac{353}{495} & -\frac{263}{495} & -\frac{142}{495} \\ -\frac{43}{495} & -\frac{362}{495} & \frac{287}{495} & \frac{133}{495} \\ \frac{56}{495} & \frac{34}{495} & -\frac{109}{495} & \frac{34}{495} \end{bmatrix}$$

(%i35) `Minv.C;`

(%o35)
$$\begin{bmatrix} 10 & 20 & 16 & 19 \\ 21 & 5 & 12 & 9 \\ 9 & 0 & 15 & 14 \\ 19 & 15 & 19 & 7 \end{bmatrix}$$

JUISTE_OPLOSSING